GeneMeow GFP Variant Design

Updated design concept and documentation with ColabFold figures and memory-safe packaging

Team: GeneMeow | Final submission: 6 sfGFP-derived amino-acid sequences

1. Objective and scoring logic

The objective was to design GFP-family protein variants with high initial fluorescence brightness and high fluorescence retention after heat treatment. The submitted sequences are amino-acid sequences. The official facility will synthesize DNA templates, express proteins in a cell-free protein synthesis system, measure Finitial, heat-treat the samples, and measure Ffinal.

The score rewards the product of initial brightness relative to WT and retention after heating. Designs must therefore preserve the GFP fold and chromophore environment rather than optimize thermal robustness alone.

```
Relative brightness = Finitial / FinitialWT
Thermal retention = Ffinal / Finitial
Comprehensive score = Relative brightness x Thermal retention
```

2. Final computational workflow

1. Use official sfGFP as the parent scaffold.
2. Train a DMS-derived brightness model on avGFP data from GFP_data.xlsx.
3. Switch from full-sequence transfer to mutation-level brightness evidence after observing saturation for sfGFP-derived sequences.
4. Generate a library of sfGFP candidates with 3-7 substitutions and hard filters.
5. Score candidates with brightness evidence, stability proxy and risk penalty.
6. Select top24 candidates as a diversified portfolio across five design families.
7. Run ColabFold AlphaFold2_batch on WT sfGFP plus top24 candidates.
8. Calculate pLDDT, pTM, chromophore pLDDT, mutation-site pLDDT and core RMSD to WT.
9. Select final 6 variants as a portfolio rather than six near-identical top-ranked candidates.
10. Create submission_GeneMeow.csv with exactly Team_Name, Seq_ID and Sequence.

3. Brightness evidence

Brightness was not treated as an experimentally proven value. A Ridge regression model with isotonic calibration was trained on avGFP DMS brightness data and reached R2 around 0.92 on the held-out test set. Because direct full-sequence transfer from avGFP to sfGFP saturated, the final ranking used mutation-level evidence rather than an absolute sfGFP brightness prediction.

H148S was used as an sfGFP-specific brightness prior. DMS-compatible substitutions such as L220V were used as brightness-safe evidence. Q69L was excluded because its avGFP-equivalent Q68L showed a negative brightness signal. N164Y was retained as a folding/packing prior with acceptable DMS-equivalent behavior.

4. Thermal-stability and folding proxy

Thermal stability was approximated with a proxy rather than directly predicted as Ffinal at 72 degrees Celsius. The proxy combined the robust sfGFP scaffold, moderate negative surface-charge shift and folding-prior mutations. Charge was approximated at neutral pH with K/R = +1, D/E = -1 and H = +0.1. Moderate negative shift was preferred over maximal supercharging to avoid excessive mutational risk.

```
Pre-ColabFold score = 0.50 x brightness evidence + 0.40 x stability proxy - risk penalty
```

5. Top24 candidate selection

The top24 set was selected from the filtered candidate library using explicit design-family diversity. The purpose was to cover several hypotheses before structural validation, because the official team score is based on the best of six submitted variants.

Design family	Count in top24
balanced_H148S_medium_charge	6
balanced_H148S_stronger_charge	6
conservative_H148S_mild_charge	5
thermal_supercharged_H148S	4
backup_without_H148S	3

6. ColabFold structural validation

WT sfGFP and the 24 selected candidates were evaluated with ColabFold AlphaFold2_batch. The run used MMseqs2 UniRef+Environmental MSA mode, five models per sequence, three recycles, no templates and no Amber relaxation. Rank_001 models were used for structural analysis. ColabFold was used as a structural sanity check and not as a direct predictor of fluorescence or thermal retention.

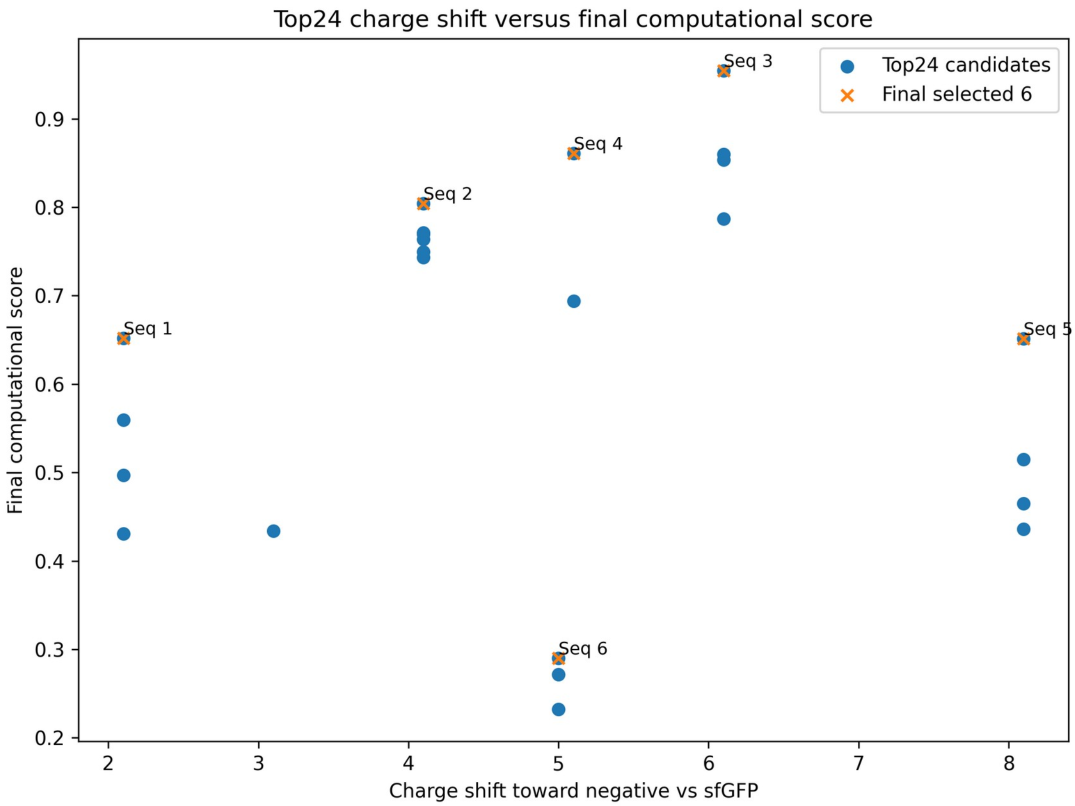

Figure 1. Top24 charge shift versus final computational score; final six candidates are highlighted.

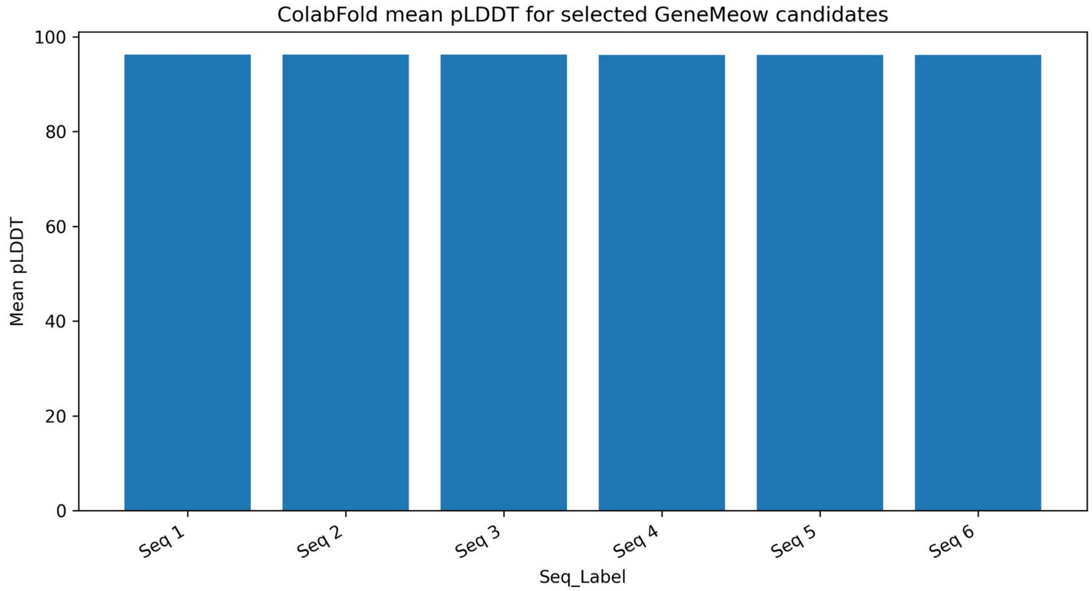

Figure 2. Mean pLDDT for the final six selected candidates.

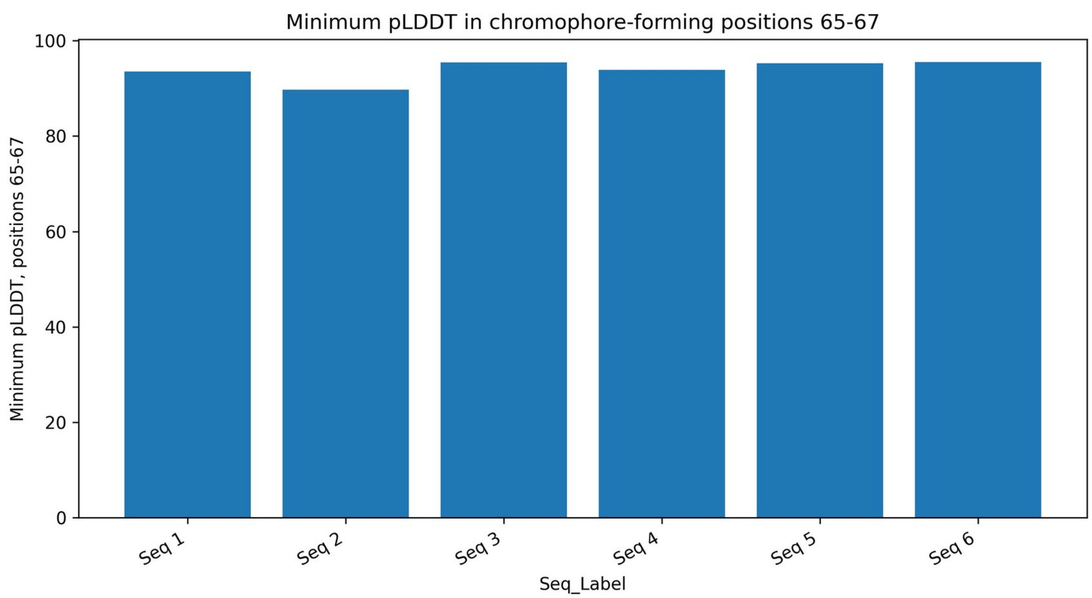

Figure 3. Minimum pLDDT in chromophore-forming positions 65-67.

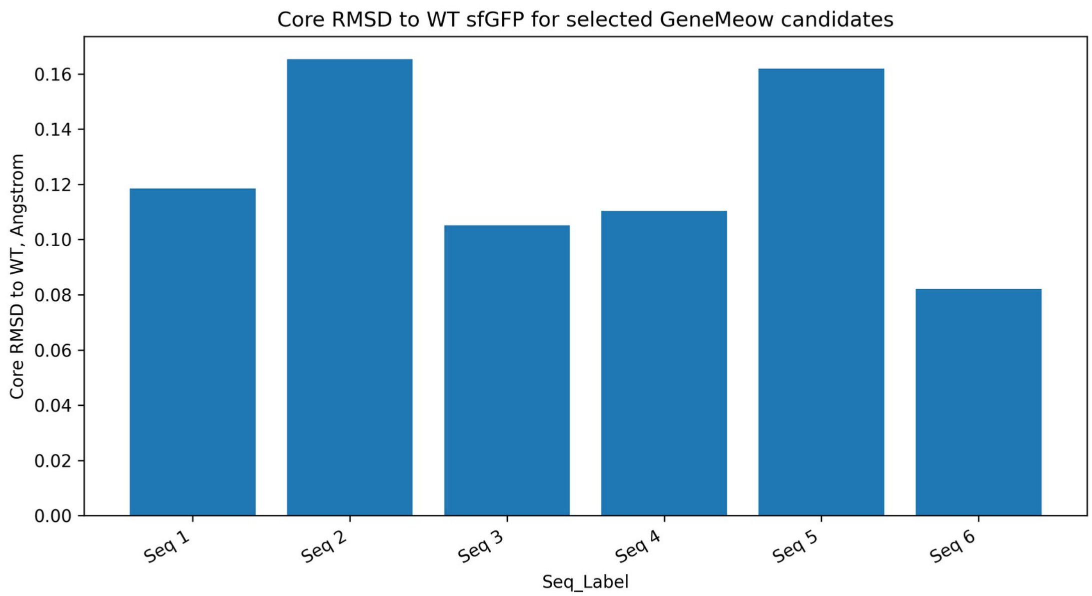

Figure 4. Core RMSD to WT sfGFP for the final six candidates.

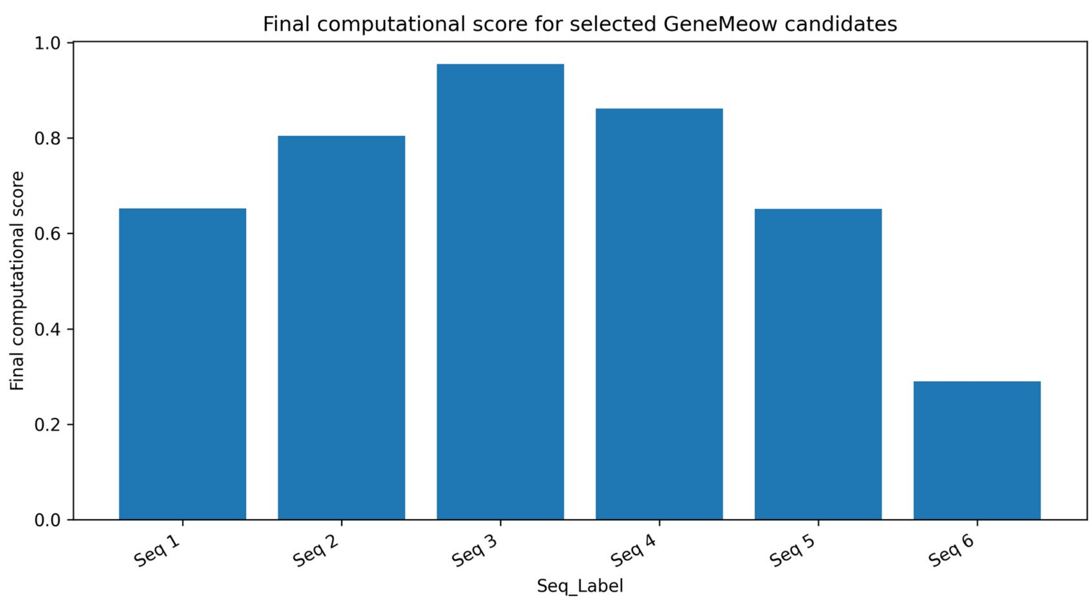

Figure 5. Final computational score for the selected candidates.

All 24 out of 24 top24 candidates passed the structural filters. The final six variants retained high mean pLDDT, high chromophore-region confidence and low core RMSD to WT sfGFP.

7. Final six selected sequences

The final set was selected as a diversified portfolio: conservative, balanced, best overall, alternative charge-pattern, thermal/supercharged bet and backup without H148S.

Seq_ID	Recommended_Role	Mutations	Mean pLDDT	Chromophore min	RMSD A	Score
1	conservative low-risk H148S + mild charge	L220V:H148S:K156E:N164Y	96.18	93.50	0.118	0.652
2	balanced medium-charge main candidate	L220V:H148S:K101E:K156E:N164Y	96.21	89.75	0.165	0.804
3	best structural/computational balanced stronger-charge candidate	L220V:H148S:K156E:K166E:K214E:N164Y	96.21	95.44	0.105	0.954
4	balanced alternative with N198D and different charge pattern	L220V:H148S:N198D:K101E:K156E:N164Y	96.11	93.88	0.110	0.861
5	thermal/supercharged high-risk high-retention bet	L220V:H148S:K101E:K156E:K166E:K214E:N164Y	96.12	95.25	0.162	0.651
6	backup without H148S	L220V:Q183R:K101E:K166E:K214E:N164Y	96.11	95.50	0.082	0.290

Team_Name	Seq_ID	Length	Starts with M	Standard AA only
GeneMeow	1	238	True	True
GeneMeow	2	238	True	True
GeneMeow	3	238	True	True
GeneMeow	4	238	True	True
GeneMeow	5	238	True	True
GeneMeow	6	238	True	True

8. Raw ColabFold plots for the main candidate

The main candidate was Query_15_CANDV2_26898 / Seq_ID 3. Representative raw ColabFold plots are shown below. Raw plots for all final six candidates are included in the GitHub-ready archive.

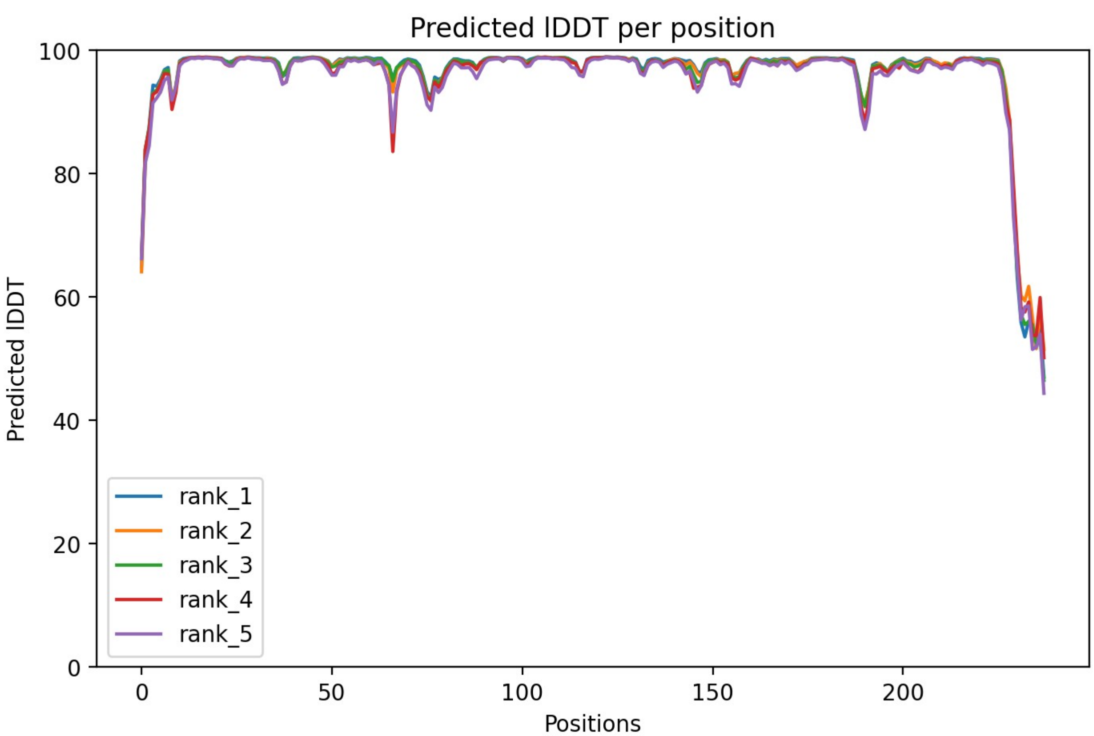

Figure 6. ColabFold pLDDT profile for Query_15 / Seq_ID 3.

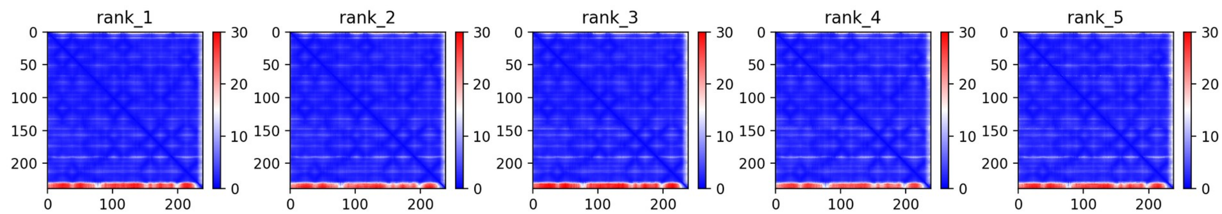

Figure 7. ColabFold PAE matrix for Query_15 / Seq_ID 3.

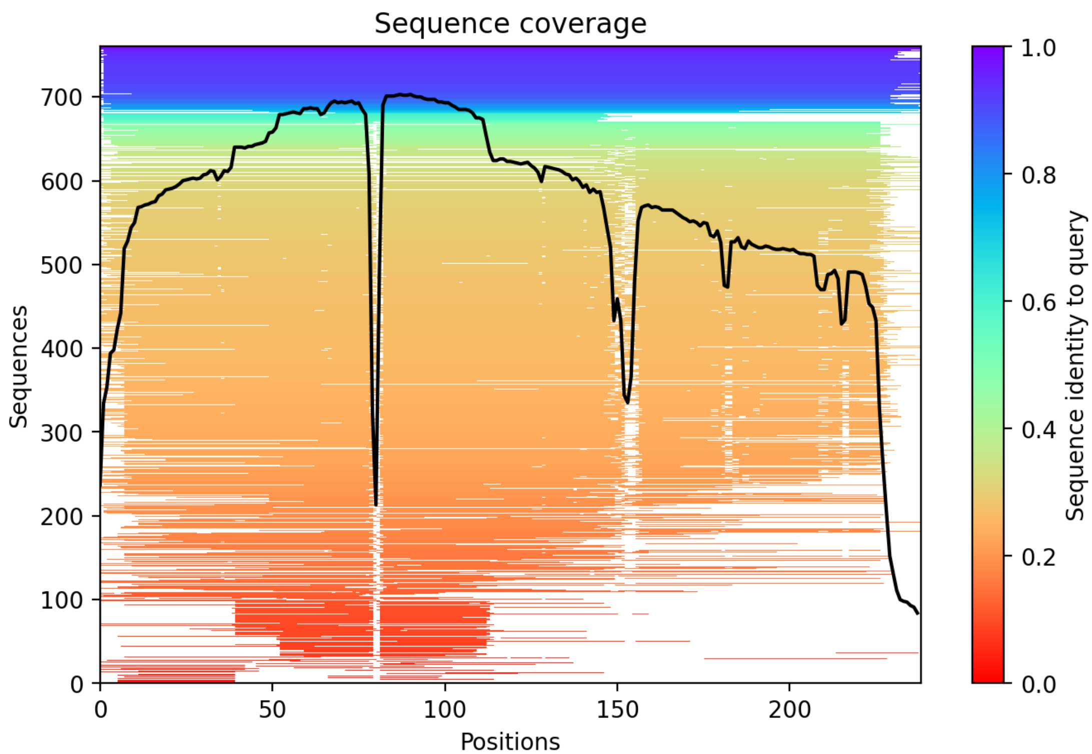

Figure 8. MSA coverage plot for Query_15 / Seq_ID 3.

9. Limitations

- The brightness score is a mutation-level prior, not an experimentally measured Finitial.
- The thermal-stability score is a charge/folding proxy, not a direct Ffinal prediction.
- ColabFold validates structural plausibility but does not predict chromophore maturation, extinction coefficient, quantum yield, CFPS yield or heat-retention score.
- Final performance can only be determined experimentally by the official facility.

10. Reproducibility and repository contents

The repository contains the design pipeline, ColabFold batch script, memory-safe final-submission script, final submission CSV, summary tables, ColabFold metrics and figures.

```
gfp-2026-genemeow-design/
  README.md
  requirements.txt
  scripts/
    gfp_2026_design_pipeline.py
    alphafold2_batch.py
    final_submission_from_colabfold_archive_memory_safe.py
  data/README.md
  outputs/
```

```
  submission_GeneMeow.csv
  final6_GeneMeow_summary.csv
  stage4_colabfold_metrics_v2.csv
  top24_for_colabfold_v2.csv
  mutation_evidence_v2.csv
figures/
  summary figures
  colabfold_raw_plots_final6/
docs/
  design_concept_GeneMeow_updated.pdf
  design_concept_GeneMeow_updated.docx
```